

# TWO CHANNELS OF DEGRADATION IN RECOMMENDER FEEDBACK LOOPS: UNIVERSAL TASTE CONTRACTION AND THE COVARIANCE SIGNATURE OF RETRAINING

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## ABSTRACT

Industrial recommender systems are periodically retrained on the interaction logs they themselves generate, closing a hidden feedback loop in which clicks cease to be a neutral measurement of preference and observed quality metrics can rise even as true utility falls. We study the resulting degradation in an agent-based simulation and on real data, and show that it decomposes into two distinct channels. The first is a universal  $\beta$ -channel: every click drags a user’s taste toward the consumed item, geometrically contracting diversity regardless of whether the model is retrained. We prove this contraction (Theorem 1) and confirm it across serving regimes and on MovieLens-20M, where it accounts for roughly a 97% loss of taste variance. The second is a retraining-specific loop channel, and its signature is subtle: we find that the commonly tracked collapse of total taste variance  $\text{tr } \hat{\Sigma}$  is driven by the  $\beta$ -channel, *not* by retraining, whereas the loop’s own fingerprint lives in the *shape* of the taste covariance and is captured by the dispersion term of  $\text{KL}(P_T \| P_0)$  rather than by the trace. This signature appears only when the model distinguishes users, the audience is multi-modal, and the population is open; we further test the natural hypothesis that retraining narrows the served distribution and refute it by direct measurement, pointing instead to retargeting of drifted users. The results yield a practical takeaway: the loop must be diagnosed through the observed–true metric gap and the geometry of the taste covariance, not through headline metrics, which improve precisely as the system degrades.

## 1 INTRODUCTION

## 2 RELATED WORK

## 3 PROBLEM SETUP

## 4 METHOD / THEORY

## 5 EXPERIMENTS

## 6 CONCLUSION

## A APPENDIX